An Analysis of Motor Vehicle Collisions in Toronto, Ontario from 2006 to 2021

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<u>Research Questions:</u>

For my final project, I decided to analyze automobile crashes in Toronto, Ontario, over 15 years and discover potential temporal trends and patterns within the urban area. Additionally, I wanted to know how these trends alter between various streets, leading to certain areas being more accident-prone than others. Finally, I will uncover which times of day are the safest and riskiest to travel on these roads and the effects of certain variables such as traffic volume, road layout of these observed patterns, and crash risk.

<u>Background</u>

Many studies have been conducted that aim to reduce and limit the risk and occurrence of motor vehicle collisions worldwide. Focusing on the study area of Toronto, Ontario, a research study completed by Fridman et al. discusses the outcome of changes in the speed limit on busy Toronto streets to determine if they reduce the overall occurrence of pedestrian-motor vehicle collisions. The city lowered the limit from 40 km/h to 30 km/h across 12 wards and used untouched streets as a means of comparison for an almost baseline for the data. The study was run from January 2015 and carried on until December 2016 to gather a large enough sample size. As a result, all streets saw a reduction in average vehicle collisions per month of just under 50%. This demonstrates a method that addresses the rising number of motor vehicle collisions while making local streets safer again (Fridman et al., 2020).

Discussing prior research outside of Canada, a study ran by Redelmeier et al., investigated the connection between cellphone use and vehicle collisions in the Boston area. Their study size consisted of 699 drivers, who were tracked based on how many phone calls they made while driving their vehicles. Overall, over 26,000 phone calls were completed over 14 months, which detailed the significant risk of collision because of distracted driving by over 400% (Redelmeier et al., 1997). This expresses another concern that is strictly related to a large share of motor vehicle collisions that can be lowered with increased education and public awareness.

Data and Study Area

I decided to focus on the city of Toronto as my study area, consisting of hundreds of major roadways that have seen motor vehicle collisions frequently. Given that Toronto is so close to London, I thought it would be appropriate to analyze one of the busiest cities in Canada. Having personal driving experience in Toronto, I believe that this analysis can help ensure traffic and pedestrian safety within the Greater Toronto Area.

I downloaded data from the Toronto Open Data Portal, where I gathered a dataset from 2006 to 2021 that detailed vehicle collisions involving fatal or severely injured people. Although it cannot be 100% accurate as to the correct number of crashes in the area, the dataset provides a good understanding of areas with increased frequency. The data is provided by the Toronto Police Services, put together through police reports and observed occurrences.

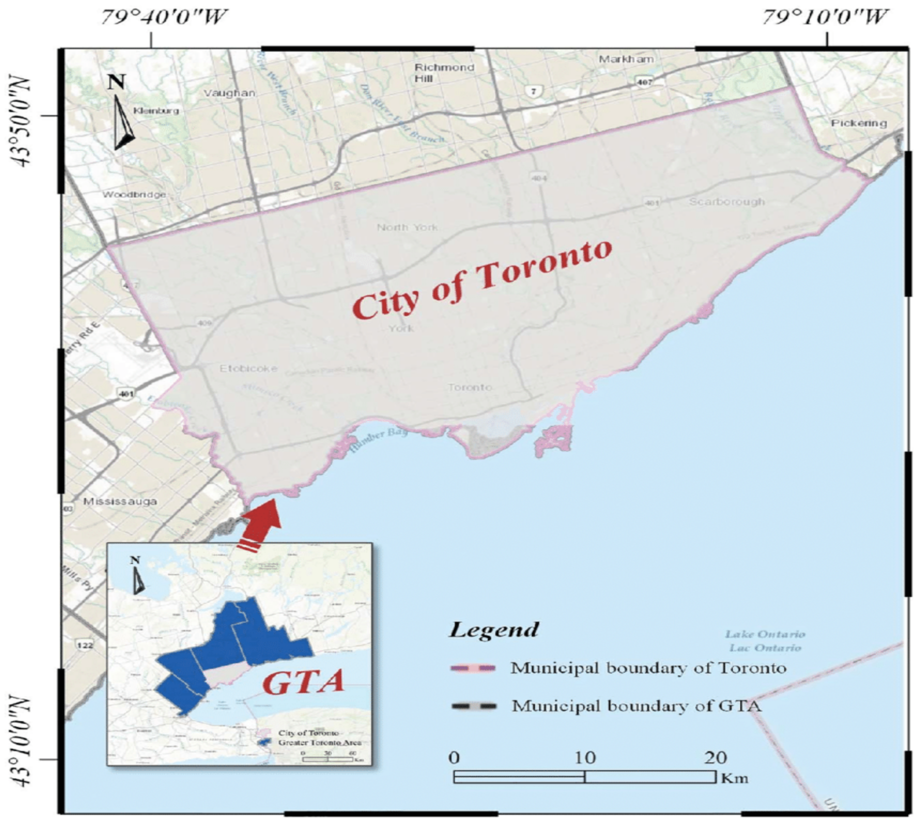

Figure 1: Study Area (Toronto, Ontario)

Analysis Procedures

Firstly, imported the dataset in the form of a shape file into the project area as a framework for future analysis tools and procedures. Figure 2 represents the base dataset, detailing the sheer prevalence of car crashes in Toronto, Ontario, from 2006 to 2021. I then used the 'Emerging Hotspot Analysis' tool to determine areas with high rates of collision. This result is detailed in Figure 3.

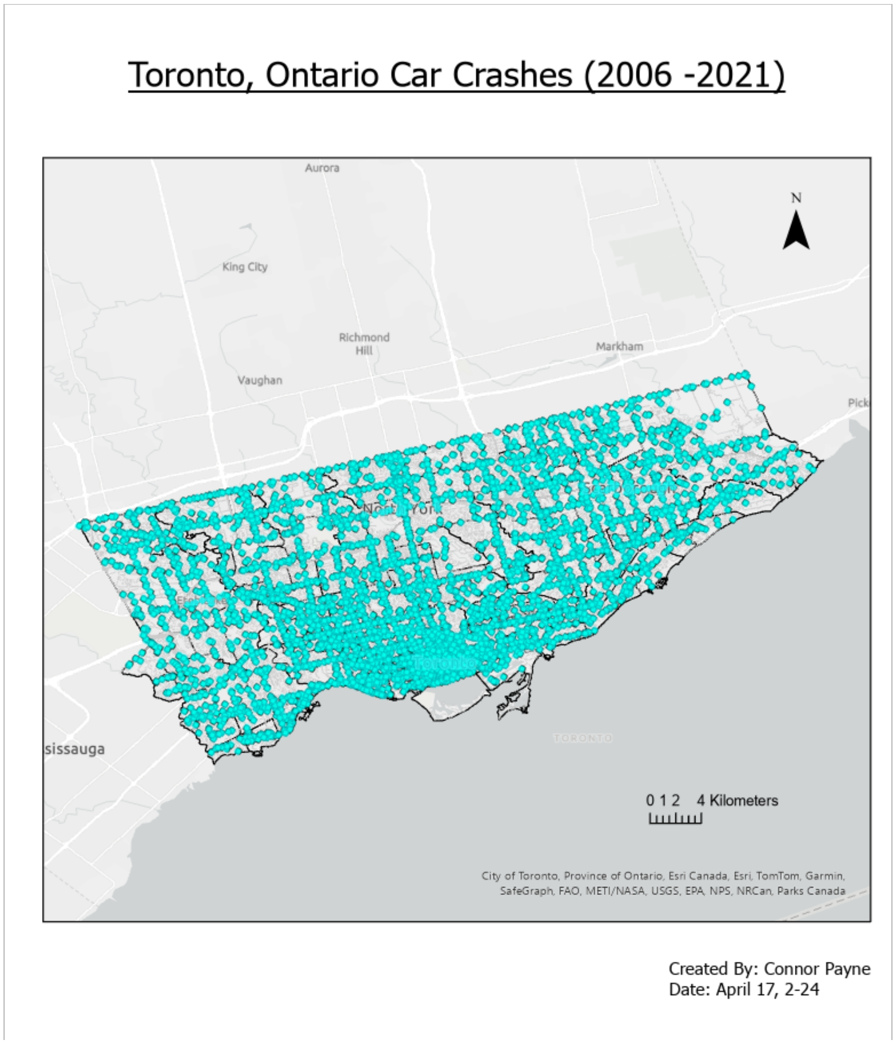

Figure 2: Base dataset from the Toronto Police Services

Additionally, I wanted to study the number of collisions that were deemed fatal within the provided data set. I copied the features from the Toronto Police Services dataset so that I still had the original layer to work from. I then went into the attribute table of the copied layer and used the 'select attributes' feature to filter out any points that were not deemed to be fatal crashes. This allowed me to visualize the severity of crashes in the Toronto area and determine the ratio between fatal and non-fatal crashes. To help me visualize these results from the above steps, I created a couple of graphs to help visualize them differently. Specifically, I wanted to discover which time of day was the most unsafe and therefore had the most crashes during those 15 years.

A line graph was created using the vehicle collision count and the time of day that the collision occurred. Overall, combining these analytical procedures has allowed me to understand and interpret the dataset more effectively and answer my research question pertaining to temporal patterns and variables associated with vehicle collision rates in Toronto, Ontario.

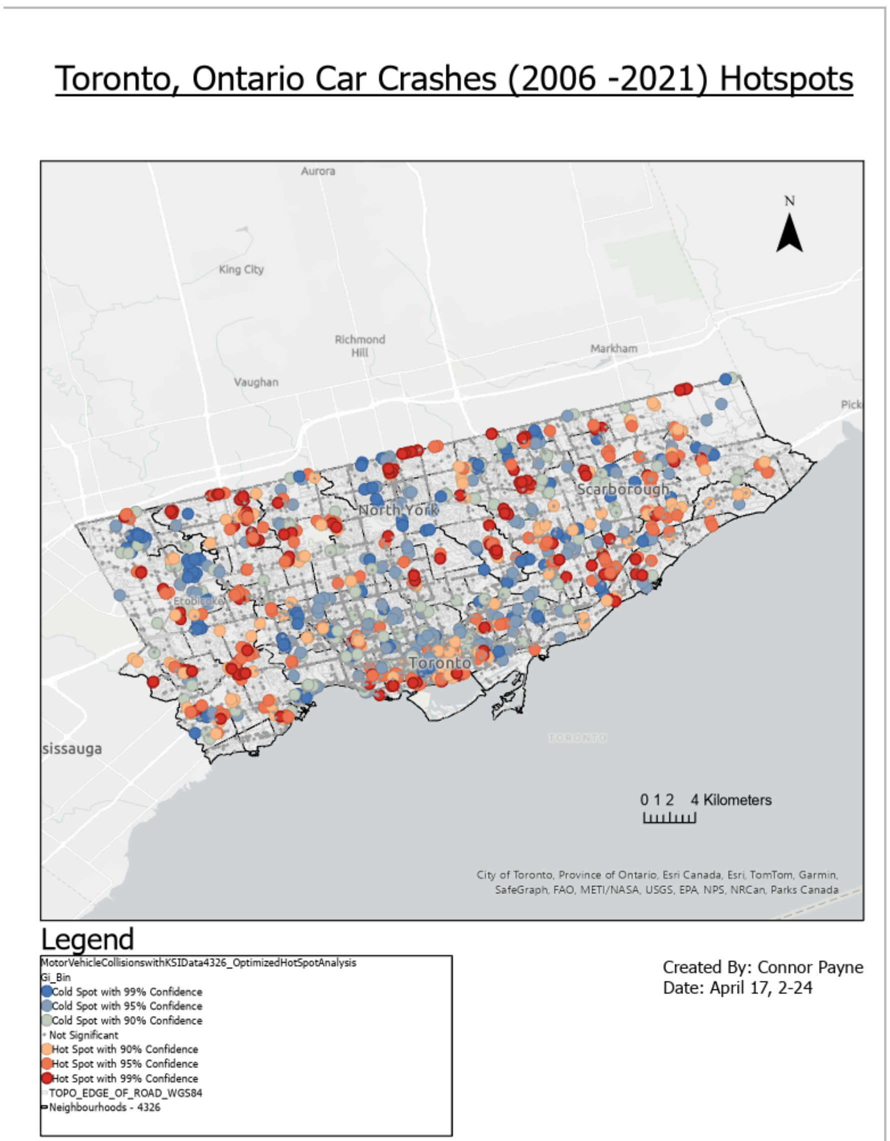

Figure 3: Emerging Hotspot Analysis Result

Results

Firstly, as we can see in Figure 2, a large majority of the city streets have experienced at least one or more crashes during the 15 years, indicating that it is a common issue in one of the busiest places in Canada. Displaying the base dataset is an effective way to interpret and compare data to determine various conditions and variables that affect the severity and overall occurrence of motor vehicle collisions. The 'Emerging Hotspot Analysis' tool allowed me to interpret areas that contain high rates of vehicle collisions and decipher why this might be. As you can see in Figure 3, there are hotspots spread across the city in some of the busier areas of Toronto. The darker red circles represent the highest rates of collisions, while the blue circles represent the lowest occurrences. The results are displayed using confidence intervals that leave room for error; however, the results are deemed to be fairly accurate when comparing the hotspot output to the base dataset. After viewing the entire hotspot output, I focused on the clusters that I saw and determined which streets they were on and why this might be. As you can see in Figure 4 (A), there are multiple clusters of hotspots around Islington Ave and Bloor St W, with additional clusters on one of Toronto's busiest streets, Dundas, and Royal York Rd. Figure 4 (B) demonstrates high volumes of clusters along the Gardner Expressway and around high tourist locations such as the Rogers Centre, Scotiabank Arena, and the CN Tower. These areas are generally extremely busy, thus leaving room for increased rates of collisions involving pedestrians and/or drivers themselves. The Gardener Expressway is one of the most congested highways, containing a high amount of traffic volume at any given time of day. This route is a popular choice for many commuting Torontonians as it connects to various other parts of the Greater Toronto Area. To determine the time of day that was the most dangerous to drive at, I created a line graph displaying the number of crashes versus the time of day, as seen in Figure 5.

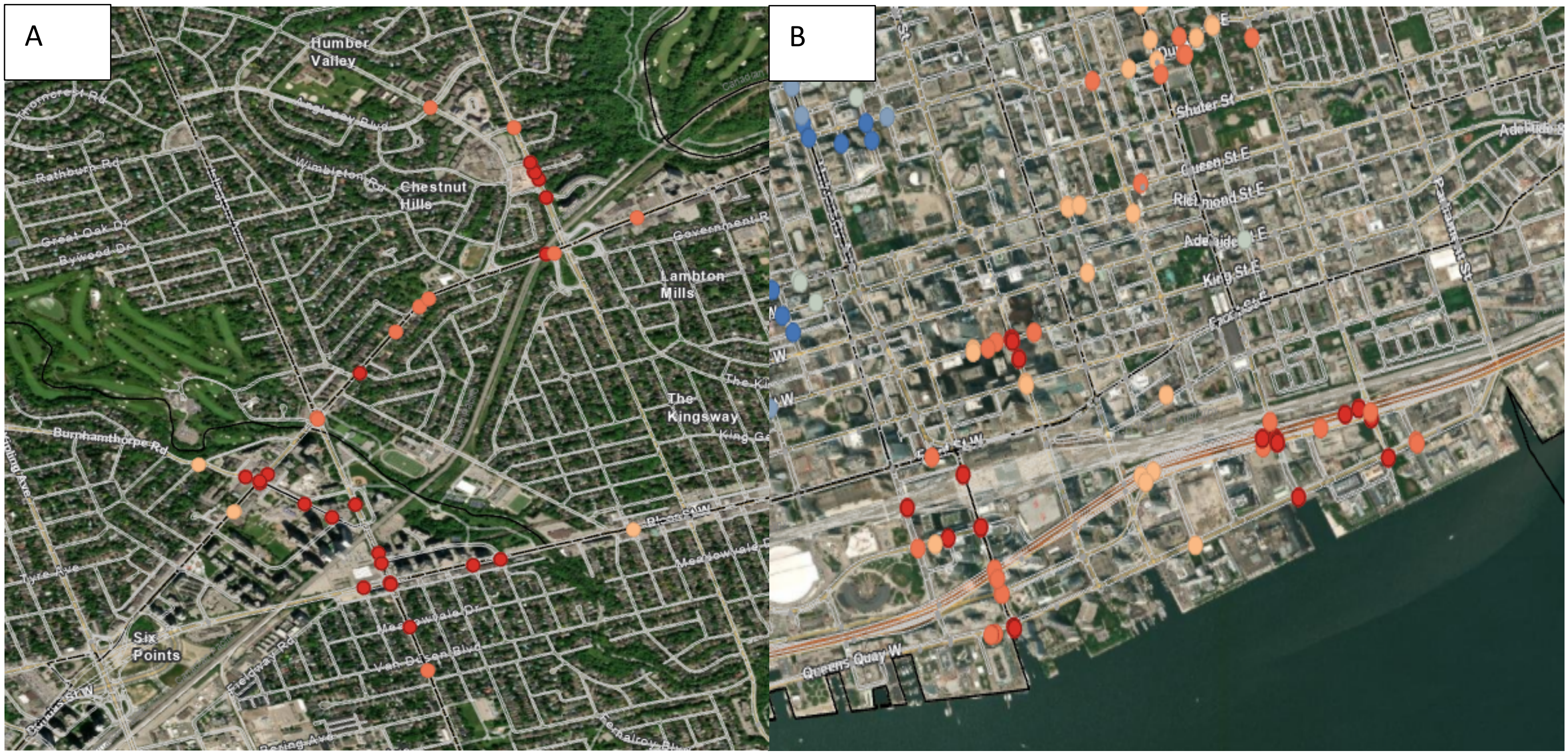

Figure 4: Hotspot Clusters

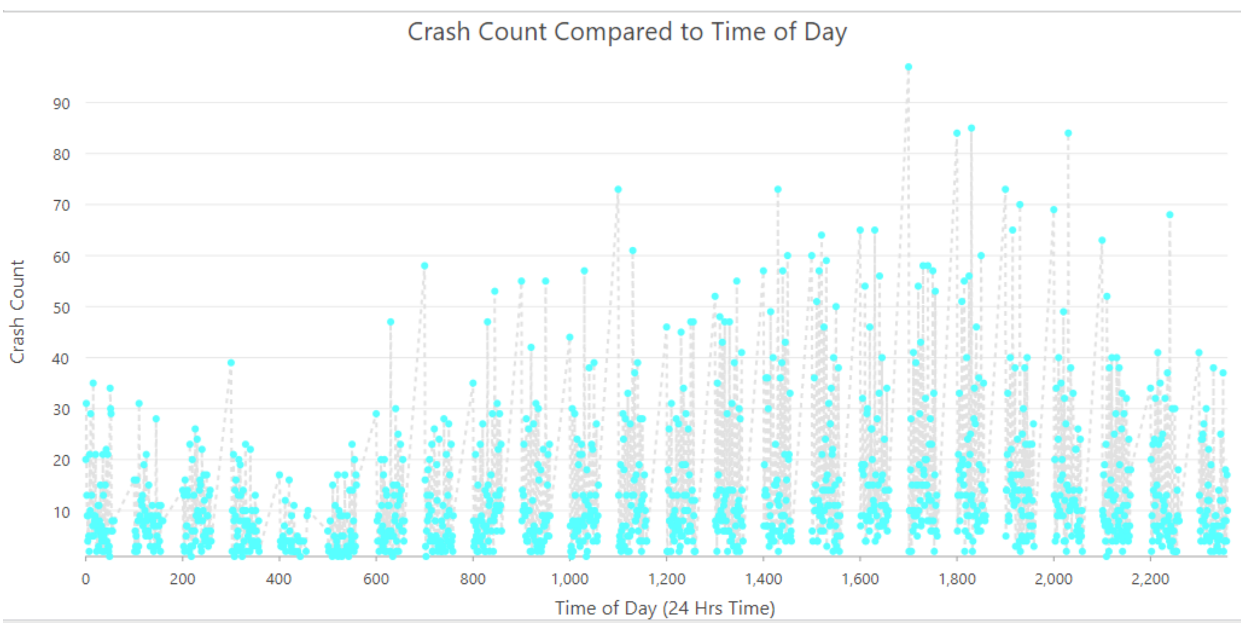

Figure 5: Crash Count versus Time of Day

We can observe when looking at the graph that there are many peak times when the roads can be the most dangerous. These include 17:00 (5 PM), 18:00 (6 PM), 20:15 (8:15 PM), and

14:15 (2:15 PM). Many of these crashes occur during times when traffic volume is at its highest, also known as 'rush hour'. Commuters want to get home as soon as possible after a long workday and are less aware of their surroundings, causing an increase in vehicle collisions. Continuing with the graphic outputs, I created a graph that displayed the Toronto neighbourhoods compared to the observed crash count to see which neighbourhoods were the highest, as seen in Figure 6. As a result, the neighbourhood with the highest crash count is Mount Olive-Silverstone-Jamestown, with 579 crashes during the reported 25 years of data. The second highest is the Yonge-Bay Corridor, with 375 observed crashes. Many neighbourhoods are within the overall average; however, these areas have the highest frequency and can be considered Toronto's most collision-heavy areas.

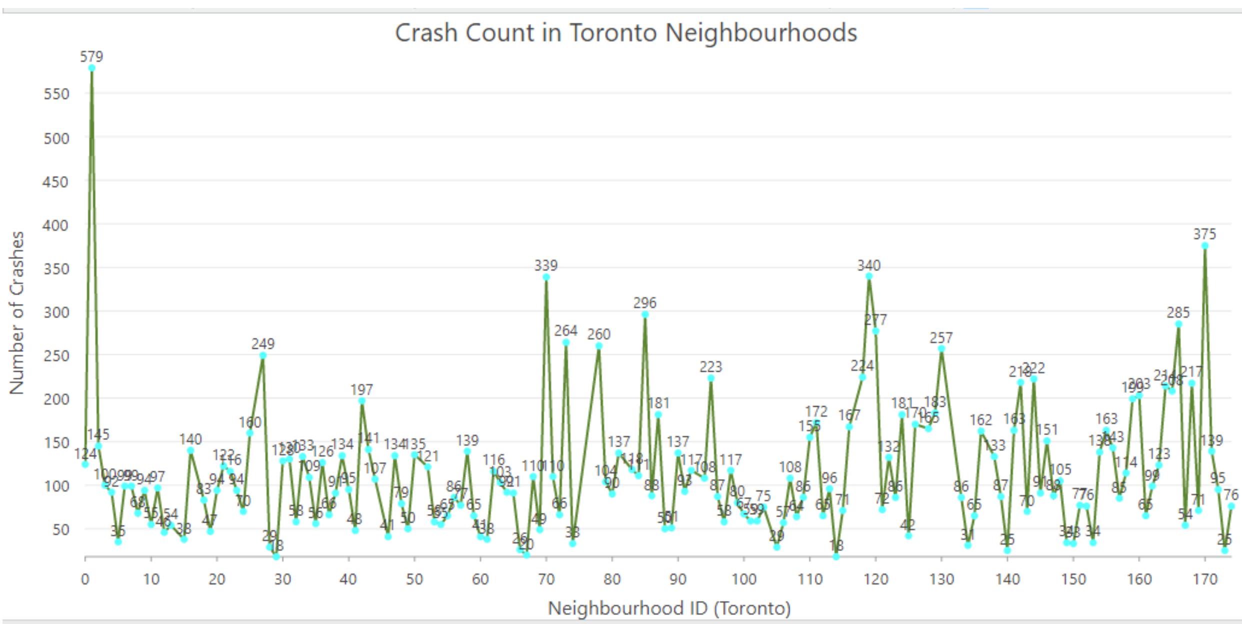

Figure 6: Crash Count vs Neighbourhoods

While conducting my analysis, I wanted to filter out the crashes that didn't result in fatalities to determine the regions of Toronto that can be observed as the deadliest when it comes to motor vehicle collisions. Figure 7, representing the overall output of the result, displays a unique comparison to the overall crash output map that still fills in the city border of Toronto.

There is a very limited difference between Figure 7 and Figure 2, which states that a majority of the crashes inside Toronto result in fatalities. I then tried to focus on specific areas that I saw had high clusters, and it was very similar to that of the hotspot analysis. Figure 8 displays large volumes along the Gardener Expressway and around the tourist destinations previously stated.

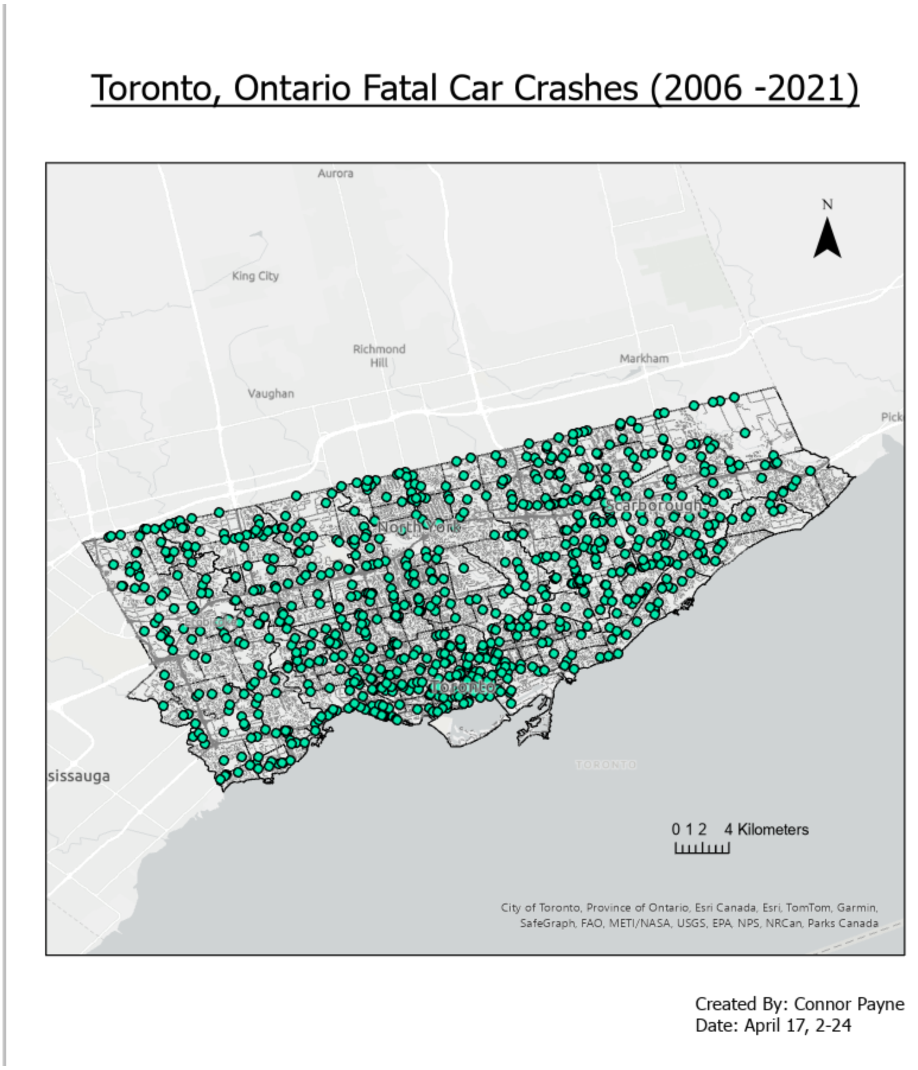

Figure 7: Toronto Fatal Car Crashes (2006-2021)

Figure 8: Fatality Clusters in Toronto

We can also observe that a large portion of the recorded fatalities caused by motor vehicle collisions are within the downtown core, where traffic is the most congested. Additionally, Figure 9 represents Yonge Street, displaying high amounts of fatalities near the intersection, causing concern for traffic safety in the area. Overall, through the various analytical approaches and procedures taken during this project, we can observe traffic patterns and areas of concern regarding motor vehicle collisions and work on ways to limit their spread.

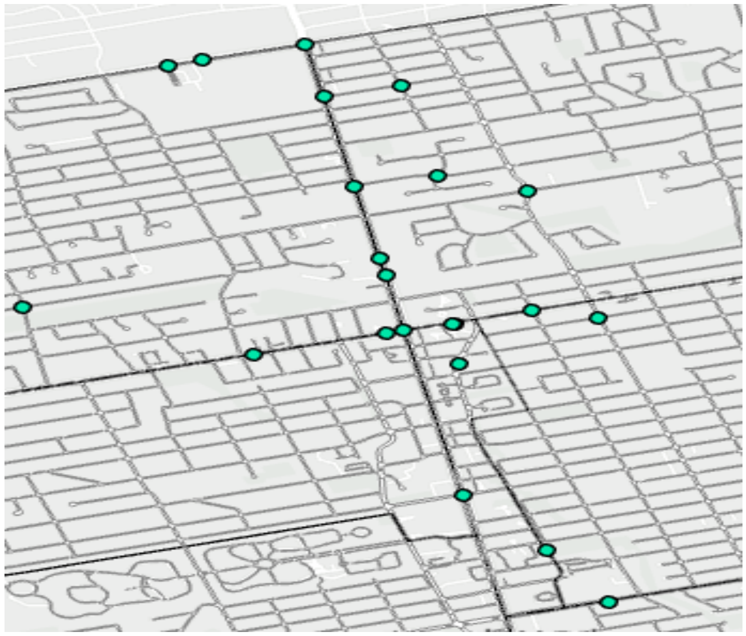

Figure 9: Yonge Street Fatal Motor Vehicle Collisions

Implications

This analysis can help provide insight into different methods of urban planning and transportation systems. Firstly, it can improve infrastructure within the city boundary involving roads, crossroads, and traffic control systems that can be designed or modified with accident prevention in mind. Installing roundabouts, pedestrian crossings, or traffic signals in accident-prone locations can help lower the risk of crashes. Additionally, city planners can help increase public safety and awareness while on the roads by encouraging safe driving practices and having local authorities create focused education and enforcement programs. This can identify the typical causes of collisions that are common, such as speeding, distracted, and drunk driving. Further, amending traffic flow by finding bottlenecks or regions of congestion that lead to accidents can be aided by the analysis of collision data. This can help avoid collisions through lane expansions, one-way street modifications, and timing adjustments for traffic lights and signals. Concerning transportation systems, citizens can be encouraged to use an alternative transport method by making adjustments, such as improving bike facilities, pedestrian infrastructure, and public transportation, that allow locals to access a safer and more environmentally friendly mode of transportation. Finally, monitoring vehicle accidents can yield valuable insights into Toronto's urban planning and transportation strategies. By using the data appropriately, planners and local authorities may work together toward creating networks that serve all populations and are more productive, safe, and sustainable.

References

Fridman, L., Ling, R., Rothman, L., Cloutier, M. S., Macarthur, C., Hagel, B., & Howard, A. (2020). Effect of reducing the posted speed limit to 30 km per hour on pedestrian motor vehicle collisions in Toronto, Canada - a quasi experimental, pre-post study. *BMC Public Health*, *20*(1). https://doi.org/10.1186/s12889-019-8139-5

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